

RM:	558843	Nome:	Laura de Oliveira Cintra
RM:	558832	Nome:	Maria Eduarda Alves da Paixão
RM:	554456	Nome:	Vinícius Saes de Souza

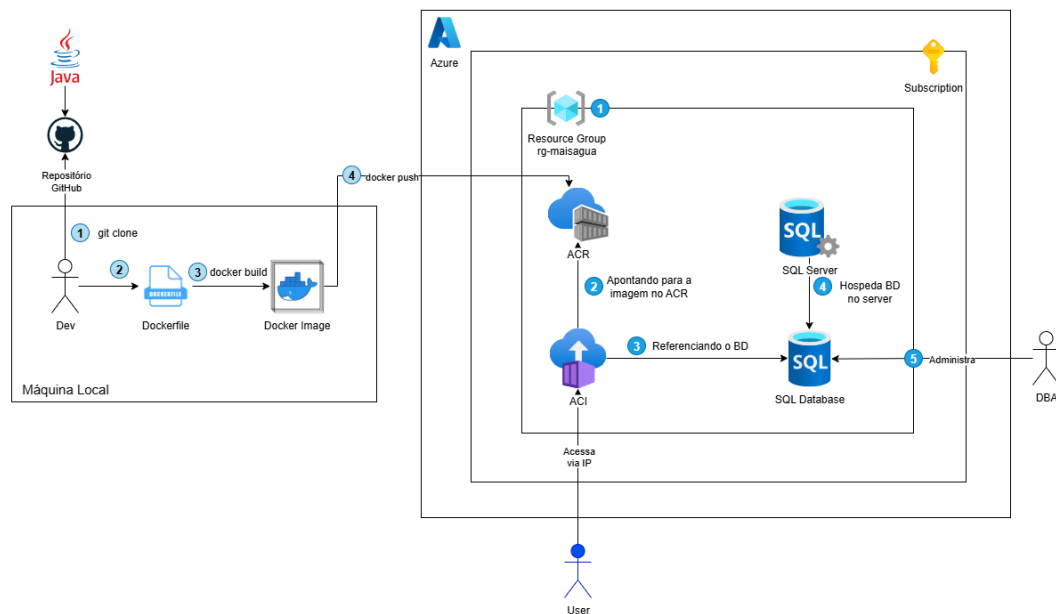
0. Link da Aplicação:

[Github](#)

1. Arquitetura da Solução

CP4 - Diagrama

RM 558843 - Laura Cintra
RM 558832 - Maria Paixão
RM 554456 - Vinícius Saes



2. Configurar um Banco de Dados no Azure SQL Server

a. Criar grupo de recursos

```
Saes@DESKTOP-1RD6S3G MINGW64 ~/Documents/programacao/Fiap/2_ano/DevOps/2_semestre/3_aula/atividade/MaisAgua-API (main)
$ az group create -l brazilsouth -n rg-maisagua
{
  "id": "/subscriptions/9443a606-9d51-4121-a77c-8bbc27b7a6ea/resourceGroups/rg-maisagua",
  "location": "brazilsouth",
  "managedBy": null,
  "name": "rg-maisagua",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
```

b. Criar Sql Server

Microsoft Azure

Search resources, services, and docs (0/17)

Copilot

RM554456@fisp.com.br
Free account. 0% reputation

Create SQL Database Server

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name *

server-maisagua

.database.windows.net

Location *

South America/Brazil/South

Authentication

Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#) >

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) or using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#) or, or select both SQL and Microsoft Entra authentication.

Authentication method

☐ Use Microsoft Entra-only authentication

☐ Use both SQL and Microsoft Entra authentication

☒ Use SQL authentication

Server admin login *

admMaisAgua

Password *

Confirm password *

c. Criar banco de dados

Create SQL Database

Microsoft

Basics

Networking

Security

Additional settings

Tags

Review + create

Product details

SQL database by Microsoft

[Terms of use](#) | [Privacy policy](#)

Estimated cost per month

5.93 USD

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription

2025-2TDSPI-Vinicius-Saes-RM554456

Resource group

rg-maisagua

Region

Brazil South

Database name

db-maisagua

Server

(new) server-maisagua

Authentication method

SQL authentication

Server admin login

admMaisAgua

Compute + storage

Basic: 1 GB storage

Backup storage redundancy

Locally-redundant backup storage

SQL

Cost summary

Basic (Basic)

Cost per DTU (in USD)

1.19

DTUs selected

x 5

ESTIMATED COST / MONTH

5.93 USD

d. SQL de criação das tabelas

Query 1 ×

 Run

☐ Cancel query

 Save query

 Export data as ⌵

 Show only Editor

```
1 CREATE TABLE tbl_pais (  
2     id_pais INT IDENTITY(1,1),  
3     nome_pais VARCHAR(50) NOT NULL,  
4     CONSTRAINT pk_pais PRIMARY KEY (id_pais),  
5     CONSTRAINT uniq_nome_pais UNIQUE (nome_pais)  
6 );  
7  
8 CREATE TABLE tbl_estado (  
9     id_estado INT IDENTITY(1,1),  
10    nome_estado VARCHAR(50) NOT NULL,  
11    id_pais INT NOT NULL,  
12    CONSTRAINT pk_estado PRIMARY KEY (id_estado),  
13    CONSTRAINT uniq_nome_estado UNIQUE (nome_estado),
```

Results

Messages

Query succeeded: Affected rows: 0

3. Configurar um ACR no Azure

Home > Container registries >



Create container registry

...

✓ Validation passed

BasicsNetworkingEncryptionTagsReview + create

Registry details

Basics

Registry name

crmaisagua

Subscription

2025-2TDSPJ-Vinicius-Saes-RM554456

Resource Group

rg-maisagua

Location

Brazil South

Domain name label scope

Unsecure

Availability zones

Disabled

Pricing plan

Basic

Role assignment permissions mode (Preview)

RBAC Registry Permissions

Networking

Public network access

Yes

Encryption

Customer-Managed Key

Disabled

Identity

None

Key Vault

None

Encryption key

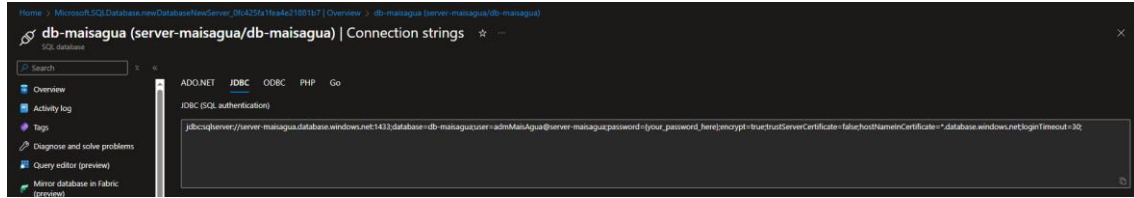
None

Version

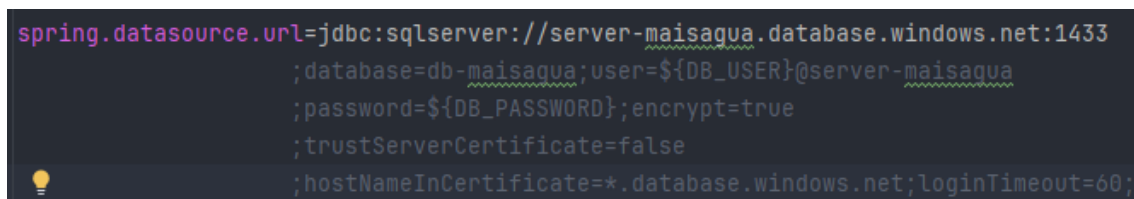
None

4. Configurar a Aplicação para Utilizar o Banco de Dados Azure SQL Server

a. Connection String do Sql server para JDBC

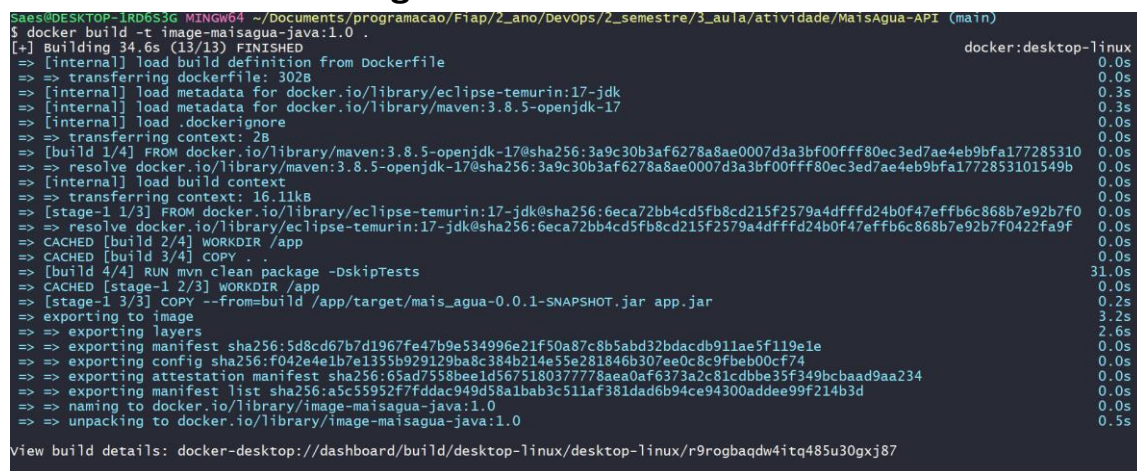


b. Application Properties do Spring



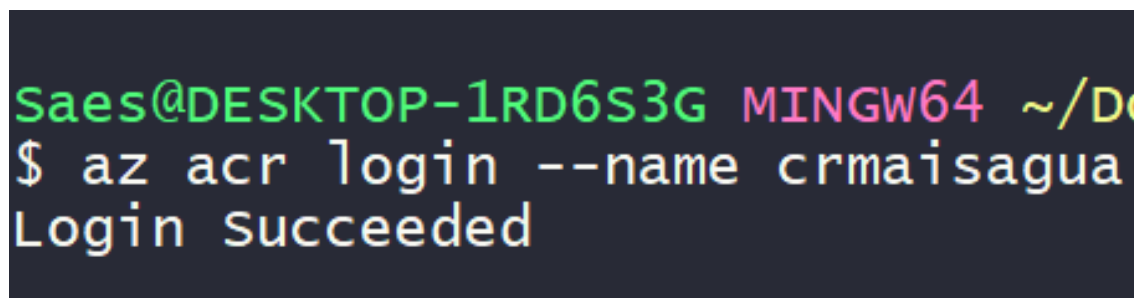
5. Criar a Imagem e Realizar o Push da Imagem no ACR

a. Criando a imagem



b. Dando push no ACR criado

i. Logando no az acr

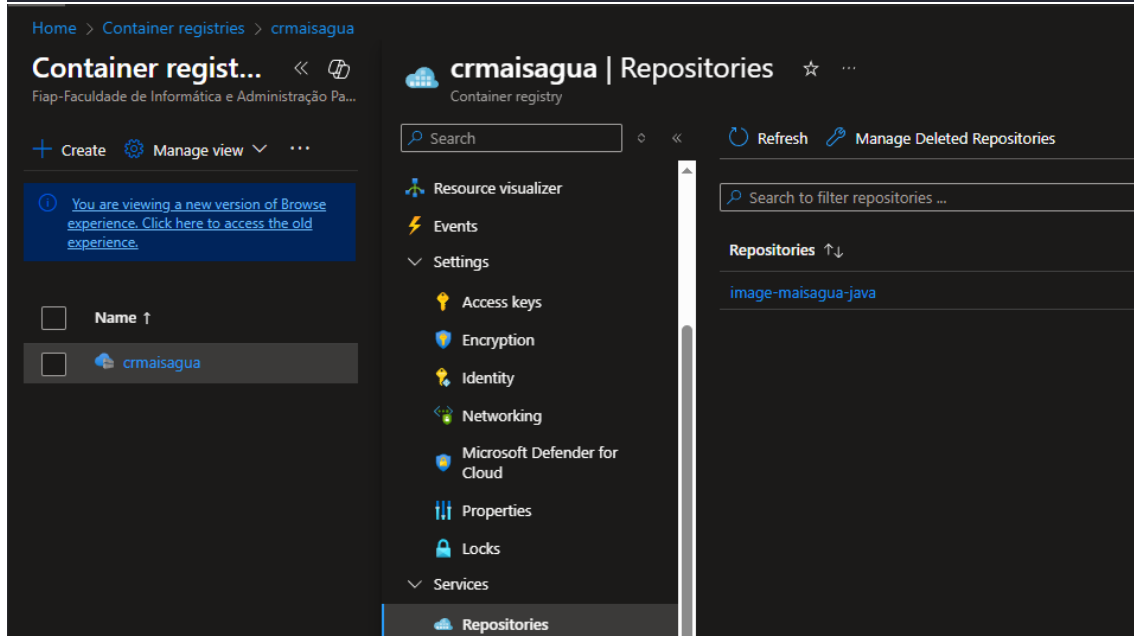


ii. Preparar tag da imagem

```
Saes@DESKTOP-1RD6S3G MINGW64 ~/Documents/programacao/Fiap/2_ano/DevOps/2_semestre/
$ docker tag image-maisagua-java:1.0 crmaisagua.azurecr.io/image-maisagua-java:1.0
```

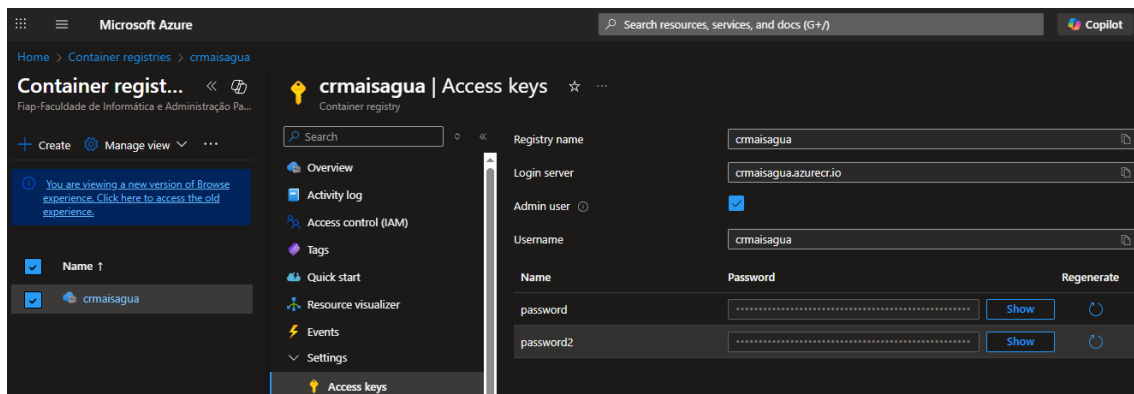
iii. Publicando no ACR

```
Saes@DESKTOP-1RD6S3G MINGW64 ~/Documents/programacao/Fiap/2_ano/DevOps/2_semestre/3_aula/ativi
$ docker push crmaisagua.azurecr.io/image-maisagua-java:1.0
The push refers to repository [crmaisagua.azurecr.io/image-maisagua-java]
17bc75678de7: Pushed
62e1f75bddef: Pushed
71f0160053e6: Pushed
18b88a7a3d70: Pushed
ced8f9c2b84a: Pushed
cdc74333da4c: Pushed
064c53302191: Pushed
b71466b94f26: Pushed
1.0: digest: sha256:a5c55952f7fddac949d58a1bab3c511af381dad6b94ce94300addee99f214b3d size: 856
```



6. Subir a Aplicação no ACI

a. Habilitando acesso admin ao Registry



b. Criando o container

Create container instance

✓ Validation passed

BasicsNetworkingMonitoringAdvancedTagsReview + create

Basics

Subscription	2025-2TDSPJ-Vinicius-Saes-RM554456
Resource group	rg-maisagua
Region	Brazil South
Container name	container-maisagua-java
SKU	Standard
Image type	Private
Image registry login server	crmaisagua.azurecr.io
Image	crmaisagua.azurecr.io/image-maisagua-java:1.0
Image registry user name	crmaisagua
OS type	Linux
Memory (GiB)	1.5
Number of CPU cores	1

Networking

Networking type	Public
Ports	8080 (TCP)
DNS name label	api-maisagua
DNS name label scope reuse	Tenant

Monitoring

Enable container instance logs	On
Monitoring workspace	DefaultWorkspace-9443a606-9d51-4121-a77c-8bbc27b7a6ea-CQ

Advanced

Restart policy	On failure
Command override	[]

Tags

(none)

 **container-maisagua-java**   

Container instances

 Search

 Start  Restart  Stop  Delete  Refresh  Give feedback

- Overview
- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Settings
- Monitoring
- Automation
- Help

 Please be aware that Docker Hub has recently introduced a pull rate limit on Docker images. When specifying an image from the Docker Hub registry, this may impact your container instance. [Learn more](#)

Essentials [JSON View](#)

Resource group [\(move\)](#) : [rg-maisagua](#)

Status : Running

Location : Brazil South

Subscription [\(move\)](#) : [2025-2TDSPJ-Vinicius-Saes-RM554456](#)

Subscription ID : 9443a606-9d51-4121-a77c-8bbc27b7a6ea

Tags [\(edit\)](#) : [Add tags](#)

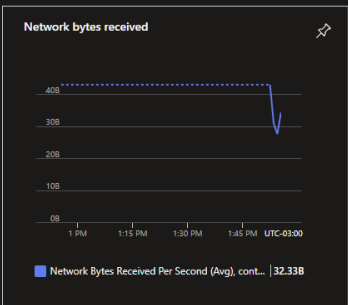
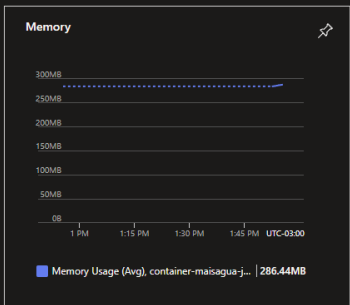
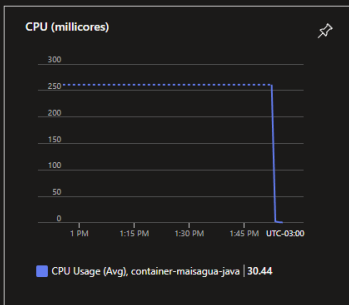
SKU : Standard

OS type : Linux

IP address (Public) : 74.163.174.181

FQDN : api-maisagua.bwe7ewbceecpehae.brazilsouth.azurecontainer.io

Container count : 1



c. Acessando o Swagger da aplicação via ACI

Swagger

/v3/api-docs Explore

+Água API - Sistema de Monitoramento Inteligente ^{v1} OAS 3.1

[/v3/api-docs](#)

API REST para gerenciamento e monitoramento inteligente de reservatórios de água com foco em eventos extremos como secas e enchentes

Servers

http://74.163.174.181:8080 - Generated server url

Histórico do Reservatório

Operações relacionadas ao monitoramento do nível de água nos reservatórios

- GET** /historico-reservatorio/{id} Buscar histórico por ID
- PUT** /historico-reservatorio/{id} Atualizar histórico
- DELETE** /historico-reservatorio/{id} Excluir histórico
- GET** /historico-reservatorio Listar históricos
- POST** /historico-reservatorio Criar histórico manualmente

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'http://4.228.74.11:8080/cadastro' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "nome": "saes",
    "email": "saes@teste.com",
    "senha": "12345"
  }'
```

Request URL

http://4.228.74.11:8080/cadastro

Server response

Code	Details
201	Response body <pre>{ "idUsuario": 1, "nome": "saes", "email": "saes@teste.com" }</pre> Response headers <pre>cache-control: no-cache,no-store,max-age=0,must-revalidate connection: keep-alive content-type: application/json date: Tue, 26 Aug 2025 17:10:52 GMT expires: 0 keep-alive: timeout=60 pragma: no-cache transfer-encoding: chunked vary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers x-content-type-options: nosniff x-frame-options: DENY x-xss-protection: 0</pre>

Home > db-maisagua (server-maisagua/db-maisagua)

db-maisagua (server-maisagua/db-maisagua) | Query editor (preview) ☆ ...

SQL database

Search Login + New Query Open query Feedback Getting started

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance
Monitoring
Automation
Help
Resource health
Support + Troubleshooting

db-maisagua (RM554456@fiap.com.br)

Showing limited object explorer here. For full capability please click here to open Azure Data Studio.

Tables

- dbo.tbl_alerta
- dbo.tbl_cidade
- dbo.tbl_clima_resumo_sem
- dbo.tbl_dispositivo
- dbo.tbl_endereco
- dbo.tbl_estado
- dbo.tbl_historico_reservatorio
- dbo.tbl_leitura_dispositivo
- dbo.tbl_pais
- dbo.tbl_reservatorio
- dbo.tbl_reservatorio_dispositivo
- dbo.tbl_status_alerta
- dbo.tbl_status_reservatorio
- dbo.tbl_unidade
- dbo.tbl_usuario

Query 1

Run Cancel query Save query Export data as Show only Editor

```
1 SELECT * FROM tbl_usuario;
```

Results Messages

Search to filter items...

id_usuario	nome	email	senha
1	saes	saes@teste.com	\$2a\$10\$1vslgahHXQLQ1Dpo3cOZvaLYuGNacM/mH1...

7. Limpando os recursos

```
Saes@DESKTOP-1RD6S3G MINGW64 ~/Documents/programacao/Fiap/
Agua-API (main)
$ az group delete --resource-group rg-maisagua
Are you sure you want to perform this operation? (y/n): y
```